Identifiability of spatial relations under geodetic datum and reference-frame loss

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Abstract

Geodetic coordinates are interpretable only through a datum or reference frame, its realization, and epoch. We ask which spatial relations remain identifiable when that metadata is lost, a question distinct from estimating coordinates in a free network. For centered three-dimensional configurations with known metric scale, complete labeled Euclidean chord distances leave a residual $O(3)$ invariance group. An independently interpreted directed axis reduces this to $O(2)$, a physically recoverable chiral witness reduces it to $SO(2)$, and an independently recoverable directed meridian removes the residual rotation. Thus an axis alone does not select signed relative longitude, and a stored orientation bit is sufficient only if its physical interpretation is independently recoverable. We test this hierarchy with seeded distance-geometry simulations, generic incomplete random frameworks, WGS84–GRS80 sensitivity, and an ITRF2020 satellite-laser-ranging demonstration. For 16-point random frameworks, the first tested edge-retention probability with observed full-local-rigidity frequency at least 0.95 was 0.6, where the estimated frequency was 0.996; this does not establish global rigidity. Changing WGS84 to GRS80 altered synthetic chord distances by at most 0.000225 m without changing directional classifications. In the ITRF2020 network, linear propagation from 2015.0 to 2025.0 changed pairwise chord distances by a maximum of 0.691 m and a median of 0.091 m. The framework identifies which archival claims are invariant, which require physical anchors, and which must remain explicitly ambiguous.

Keywords: geodetic datum; reference frame; identifiability; distance geometry; rigidity; epoch

1 Introduction

A geodetic coordinate tuple is not a physical location by itself. Its interpretation depends on a coordinate reference system, datum or reference-frame realization, coordinate epoch, and coordinate operations (ISO 2019; Petit and Luzum 2010). Terrestrial reference frames realize origin, scale, orientation, and time evolution from observations and conventions (Altamimi et al. 2002, 2023). Numeric coordinates and trajectory models remain epoch- and model-dependent (Bevis and Brown 2014), yet can outlive the metadata, station realization, software, or object identifiers needed to interpret them.

Datum defects, rank-deficient normal equations, free-network adjustment, inner constraints, and minimum constraints are established geodetic topics (Papo 1985; Sillard and Boucher 2001; Kotsakis 2012, 2013; Kotsakis and Chatzinikos 2017). Coordinate-invariant network models clarify which parameters are non-

estimable before a datum is imposed (Chatzinikos and Dermanis 2017), while no-net-rotation and station-selection choices affect realized frames (Altamimi et al. 2003; Glaser et al. 2015; Zajdel et al. 2019).

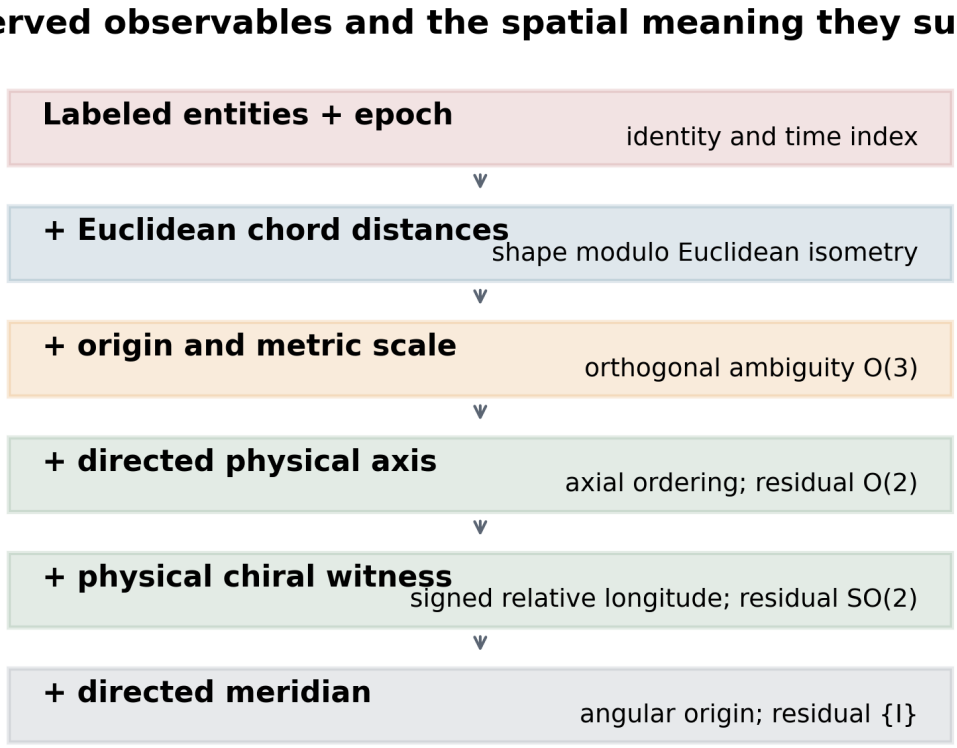


Fig. 1 Preserved observables and the spatial meaning they support. Each layer adds information not recoverable from the preceding layer under the stated Euclidean model.

Our question begins after that operational setting has failed: given only selected archived observables, which relational claims remain identifiable, and what residual invariance group remains? We focus on labeled Euclidean chord distances, axial ordering, signed relative longitude, and absolute angular origin. Fig. 1 separates the information layers. This is a sufficiency audit for archival observables, not a new datum-adjustment method.

The contribution is threefold: a corrected residual-invariance hierarchy that separates axis, physical chirality, and meridian information; explicit definitions and degeneracy states for directional relations; and a reproducible analysis of incomplete-network local rigidity, ellipsoid sensitivity, and epoch dependence in an ITRF2020 example. The synthetic reflection and corruption experiments serve as theorem validation rather than as empirical discoveries.

2 Geodetic observables and terminology

We follow IERS usage: a terrestrial reference system is the theoretical specification, whereas a terrestrial reference frame is its realization by coordinates and motions of physical points (Petit and Luzum 2010; Altamimi et al. 2023). A datum, coordinate reference system, and coordinate operation are metadata needed

to interpret coordinate values, but none is identical to the physical relation being archived. An ellipsoid such as GRS80 specifies shape and gravity-related constants, not a complete time-evolving frame (Moritz 2000).

The theoretical distance is $d_{ij} = \| x_i - x_j \|_2$ in a three-dimensional Euclidean embedding: a chord distance, not a surface geodesic on a sphere or ellipsoid. Surface-geodesic distance defines a different distance-geometry problem and is outside the present theorems.

For a nonzero directed axis s and nonzero point x measured from a specified origin, define generic axial latitude as $\phi_s(x) = \arcsin \frac{s \cdot x}{\| s \| \| x \|}$. This becomes geocentric latitude only when the origin and directed axis have the corresponding terrestrial interpretation. It is not geodetic latitude, which uses the ellipsoid normal; astronomical latitude, which uses the local gravity direction; or magnetic latitude, which uses a magnetic-field model. We use north/south only when the directed axis has an independently defined physical interpretation.

For projections $a_\perp$ and $b_\perp$ onto the plane normal to s, define signed relative longitude by $atan2[s \cdot (b_\perp \times a_\perp), a_\perp \cdot b_\perp]$ on the principal branch $-\pi < \lambda \leq \pi$. The relation is pairwise and is not a globally transitive total order. It is reported as undefined for axis projections and exact or near projected antipodes; same-meridian pairs are explicit ties. Branch-cut proximity is therefore a conditioning issue, not hidden by a forced east/west label.

We represent an archived relation event as $E = \{ I, t_{obs}, t_{ref}, G, O, C, M, U, P \}$. I stores entity identifiers and identity hypotheses; t_{obs} and t_{ref} are observation and reference epochs; G is relational geometry; O contains origin and directed-axis information; C is a physically interpretable chiral witness; M is a directed meridian or equivalent angular anchor; U is uncertainty; and P records provenance and model assumptions. Entity identity and epoch recoverability are independent of geometric precision.

3 Identifiability theory

Let $X = [x_1, \ldots, x_n]^\mathsf{T}$ be labeled points in $\mathbb{R}^3$ and let D contain $d_{ij} = \| x_i - x_j \|$. Double centering of squared distances gives $B = -\frac{1}{2} J (D \circ D) J$, where $J = I - 1 1^\mathsf{T} / n$. An eigendecomposition of B recovers centered coordinates whenever D is Euclidean. For a generic rank-three configuration, any other realization with the same D is $X Q + 1 c^\mathsf{T}$ for orthogonal Q and translation c. If physical units are not stored, a scale factor is also free (Dokmanić et al. 2015; Liberti et al. 2014).

For a retained observable map F, we call the transformations Q for which $F(QX) = F(X)$ the residual invariance group of those observables. This is an observational equivalence group, not the pointwise stabilizer of a generic realized configuration.

Proposition 1 (metric identifiability). For a generic complete labeled Euclidean distance matrix of embedding dimension three, the realization is unique only modulo translation and an element of $O(3)$. Proof sketch: double centering determines the centered Gram matrix. Any rank-three factorization of that Gram matrix differs by an orthogonal factor. Because $O(3)$ contains both proper and improper transformations, distance data do not select chirality.

After translation is removed and metric scale is known, the residual invariance group is $O(3)$. Requiring an independently interpreted directed axis s leaves an $O(2)$ residual invariance group because axial reflections remain. A physically recoverable chiral witness selects the proper branch and leaves $SO(2)$, the rotations about s. An independently recoverable directed meridian removes that final rotation. Fig. 2 and Table 1 summarize the chain $O(3) \to O(2) \to SO(2) \to \{I\}$.

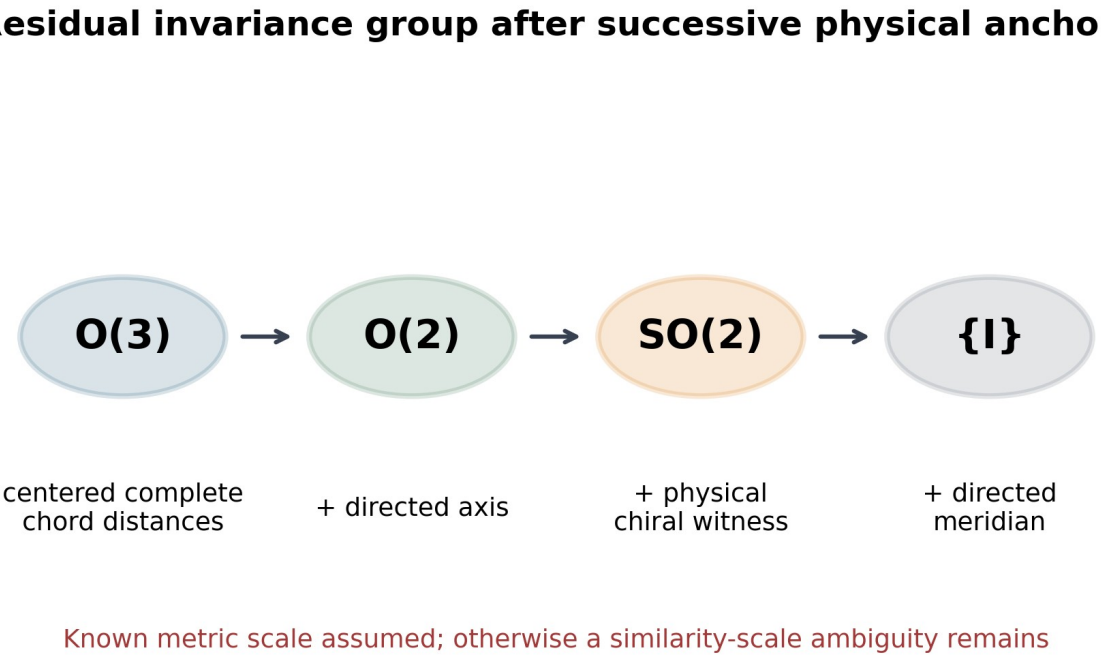


Fig. 2 Residual invariance group after successive physical anchors. Complete centered Euclidean chord distances leave O(3); a directed axis leaves O(2); an independently recoverable physical chiral witness leaves SO(2); and a directed meridian removes the final axial rotation.

Table 1 Information hierarchy and residual ambiguity under a rigid model.

Model	Stored Information	Identifiable Quantities	Remaining Ambiguity
M0	labeled entities only	entity identity only	unconstrained geometry
M1	+ selected chord distances	recorded edge lengths	graph-dependent
M2	+ locally rigid distance graph with known metric scale	local Euclidean shape	Euclidean isometry E(3); possibly discrete noncongruent embeddings
M3	+ complete labeled Euclidean chord-distance matrix	global Euclidean shape for a generic full-dimensional configuration	translation and O(3)
M4	+ fixed origin or centered coordinates	origin-referenced shape	O(3)
M5	+ independently interpreted	axial latitude ordering	residual invariance group O(2)

Model	Stored Information	Identifiable Quantities	Remaining Ambiguity
	directed axis		
M6	+ physically recoverable chiral witness	signed relative longitude away from degeneracies	proper axial rotations SO(2)
M7	+ independently recoverable directed meridian	absolute angular origin about the axis	identity
Metadata	+ observation epoch, entity continuity, units, and model provenance	time-indexed physical interpretation	model and measurement uncertainty remain

Proposition 2 (axial-order invariance). Let Q be any orthogonal transformation applied jointly to centered points and directed axis. Because $(Qs)\cdot(Qx)=s\cdot x$ and norms are preserved, axial latitude and its pairwise ordering are invariant under both proper and improper transformations. The directed sense is essential: replacing s by -s exchanges the ordering.

Proposition 3 (chirality dependence). Under the same joint orthogonal transformation, the scalar triple product defining the sine term is multiplied by $det(Q)$, whereas the cosine term is unchanged. Signed relative longitude is therefore invariant under $SO(3)$ and reverses under an axial reflection in $O(3)$ $SO(3)$, as illustrated in Fig. 3. An algebraic orientation label resolves this ambiguity only if the physical interpretation of the label is independently recoverable; a coordinate-handedness convention alone is not a physical witness.

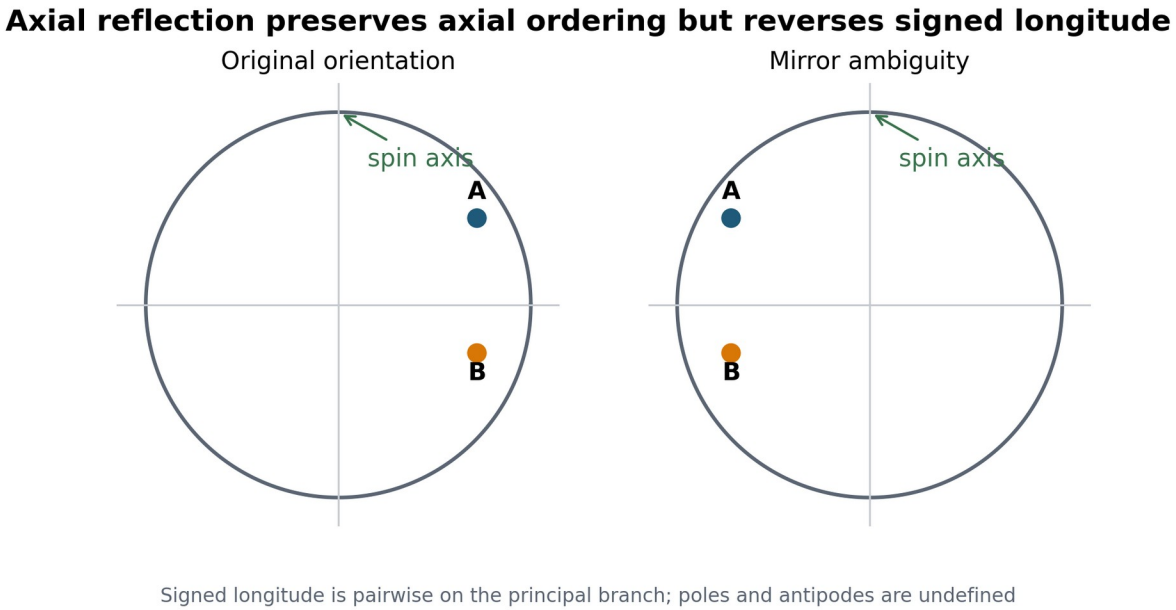


Fig. 3 Directional asymmetry under axial reflection. Generic axial-latitude ordering is preserved, whereas signed relative longitude reverses. The longitude relation is pairwise on the principal branch and is undefined at axis and projected-antipode degeneracies.

Proposition 4 (meridian necessity). With origin, directed axis, and physical chirality fixed, every proper rotation about s preserves distances, axial ordering, and signed relative longitudes. These observables cannot

identify an absolute angular origin. A directed meridian or equivalent oriented external direction at the relevant epoch is required. Without physical chirality, axis plus meridian still leaves the reflection fixing both.

These statements concern identifiability, not necessarily stable estimation. Identifiability asks whether two admissible states can produce exactly the same retained information; conditioning asks how strongly noise in that information changes an estimate. A nearly coplanar oriented simplex may formally encode chirality while being numerically fragile. The archive must therefore store uncertainty and quality indicators for anchors rather than only a categorical claim that an anchor exists.

For incomplete distance graphs, local rigidity of a generic three-dimensional framework requires rigidity-matrix rank $3n - 6$ (Asimow and Roth 1978, 1979). This rank condition removes the six infinitesimal Euclidean gauge motions but does not establish global rigidity or exclude disconnected noncongruent realizations. It is therefore used here as a limited network-identifiability diagnostic.

4 Materials and methods

The synthetic benchmark samples 24 points on a WGS84-shaped ellipsoid, forms the complete distance matrix, and reconstructs a three-dimensional configuration by classical multidimensional scaling. Proper and reflection-permitting Procrustes fits are computed independently from the same unaligned reconstruction using the determinant constraint of Kabsch (1976, 1978).

250 random proper rotations and axial reflections validate the directional invariance claims. 4000 Monte Carlo trials per condition perturb the axis and corrupt an algebraic orientation label; these are symmetry sanity checks, not models of physical witness survival. Wilson intervals quantify finite-simulation uncertainty. The frame taxonomy in Table 2 distinguishes the physical meaning and failure modes of candidate anchors.

Table 2 Reference-frame classes, physical anchors, and principal failure modes.

Frame	Physical Anchor	Preserving Transformations	Temporal Durability	Principal Failure Mode	North South Support	East West Support
Terrestrial reference frame	realized station coordinates and velocities	time-dependent Euclidean/Helmert mapping	high for modern epochs	network/model discontinuity	yes	yes
Celestial reference frame	extragalactic source directions	celestial rotations	high	source structure and realization change	with Earth orientation	with Earth orientation
Spin-axis frame	directed angular-momentum/rotation axis	rotation about axis and axial reflection	medium	polar motion and true polar wander	yes	only with physical chirality
Plate-fixed frame	rigid plate model	plate Euler rotation	medium	deformation and plate reorganization	model-dependent	model-dependent
Paleomagnetic frame	remanent field direction under field assumptions	axial rotations	geological	remagnetization and GAD violations	paleolatitude	not from paleomagnetism alone
Hotspot/mantle frame	hotspot tracks or mantle structures	model-specific plate rotations	geological	mantle motion and sparse anchors	model-dependent	model-dependent
Relative tectonic network	distances/rotations among blocks	network gauge transformations	geological	topology loss and deformation	only with axial anchor	only with orientation anchor
Hybrid redundant frame	independent geometric, axial, and physical anchors	intersection of preserved symmetries	highest conditional robustness	correlated anchor failure	yes	yes

The incomplete-network experiment draws generic points from a continuous three-dimensional distribution for n = 6, 8, 10, 12, 16. Every possible edge is retained independently with probability in {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0}, with 2000 point-and-graph trials per cell. We calculate the numerical rigidity-matrix rank, full-local-rigidity frequency, Wilson interval, and excess nullity after the six Euclidean gauge modes. Table 3 reports the first tested edge-retention probabilities with observed local-rigidity frequencies at or above 0.50 and 0.95; it is a seeded phase diagram, not a universal threshold.

Table 3 First tested edge-retention probabilities with observed local-rigidity frequencies at or above 0.50 and 0.95 in the seeded generic-random-framework experiment; these are not universal thresholds.

N Points	First Tested Edge Probability At Or Above 50Pct	First Tested Edge Probability At Or Above 95Pct	Complete Graph Local Rigidity Probability
6	0.8	1.0	1.0
8	0.7	0.9	1.0
10	0.6	0.7	1.0
12	0.5	0.7	1.0
16	0.4	0.6	1.0

Ellipsoid sensitivity compares WGS84 with GRS80 using identical seeded geodetic latitudes and longitudes. We report coordinate and chord-distance differences and agreement of axial-order and signed-longitude classifications. This isolates dependence on the reference ellipsoid from the group-theoretic claims.

For a realized geodetic demonstration, we parse the official ITRF2020 SLR station file at reference epoch 2015.0 (Altamimi et al. 2023). Stations with maximum Cartesian coordinate standard uncertainty no greater than 0.02 m are divided into longitude bins; the best station in each occupied bin is selected and supplemented from remaining eligible stations to obtain 16 sites, listed in Supplementary Table S1. Positions are linearly propagated to 2025.0 using published velocities. This propagation is an epoch-sensitivity demonstration, not a complete nonlinear station-motion model.

Epoch sensitivity can be written schematically as $x(t)=x(t_{ref})+v(t-t_{ref})+d(t)$, where v is a secular velocity and d contains discontinuities, seasonal signals, postseismic deformation, and other nonlinear terms. Dropping t_{ref} or the motion model makes a coordinate internally incomplete. Our 10-year linear calculation retains only the secular term so that the role of epoch is visible without conflating it with a station-specific geophysical model.

The WGS84–GRS80 sensitivity values are reported in Supplementary Table S2. All reported numerical values are loaded from the generated result registry rather than copied into the document builder; the registry is reproduced as Supplementary Table S3. Calculations use seed 20260924. Raw-input checksums, acquisition metadata, result traceability, code, tests, editable tables, and editable figure slides accompany the submission.

5 Results

The incomplete-network analysis produced a size-dependent transition in local rigidity (Fig. 4). For n = 16, full local rigidity occurred in 0.947 of trials at edge-retention probability 0.5 (95% Wilson interval 0.936–

0.956) and 0.996 at 0.6. The latter 95% Wilson interval was 0.992–0.998. The first tested probability with an observed frequency at least 0.95 was therefore 0.6. Smaller frameworks required higher tested retention probabilities (Table 3). These frequencies characterize the configured random framework ensemble and do not imply global rigidity.

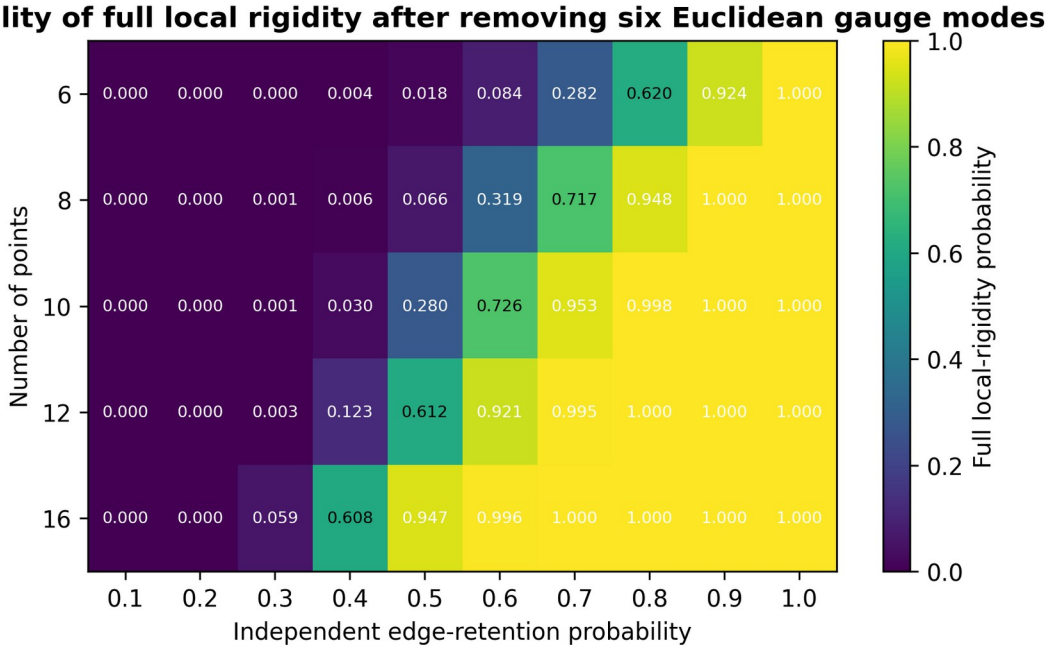


Fig. 4 Probability of full local rigidity for generic three-dimensional random frameworks. Each cell summarizes 2000 independently sampled point-and-edge realizations after the six infinitesimal Euclidean gauge modes are removed. The result concerns local rigidity, not global uniqueness.

Replacing WGS84 by GRS80 changed synthetic coordinates by at most 0.000121 m and Euclidean chord distances by at most 0.000225 m. Both axial-order and signed-longitude classifications agreed for all tested pairs. The sensitivity values are reported in Supplementary Table S2.

The complete distance matrix reconstructed the synthetic configuration with a reflection-permitting RMSE of 2.97e-09 m. For the seeded realization, constraining the alignment to a proper rotation exposed an RMSE of 6.136 million m, demonstrating that a numerically exact metric recovery need not recover handed orientation.

Across random proper rotations, north/south and east/west sign accuracies were both 1.000. Under axial reflection, north/south accuracy remained 1.000, whereas east/west agreement was 0.000. These validation results are shown in Supplementary Fig. S1.

The Monte Carlo sanity check separated axis noise from algebraic orientation-label corruption. At zero axis noise and label-corruption probabilities of 0.1 and 0.4, north/south accuracy remained 1.000; east/west accuracy decreased to 0.895 (95% Wilson interval 0.885–0.904) and 0.602 (95% Wilson interval 0.587–0.617), respectively (Supplementary Fig. S2).

The ITRF2020 demonstration selected 16 SLR stations (Fig. 5). Distance reconstruction again had negligible reflection-permitting RMSE (3.61e-09 m). A computational reflection preserved all axial-order signs and

reversed all nondegenerate signed-longitude signs; it is a symmetry test, not a physically valid ITRF realization. From 2015.0 to 2025.0, the maximum absolute pairwise-distance change was 0.691 m and the median was 0.091 m.

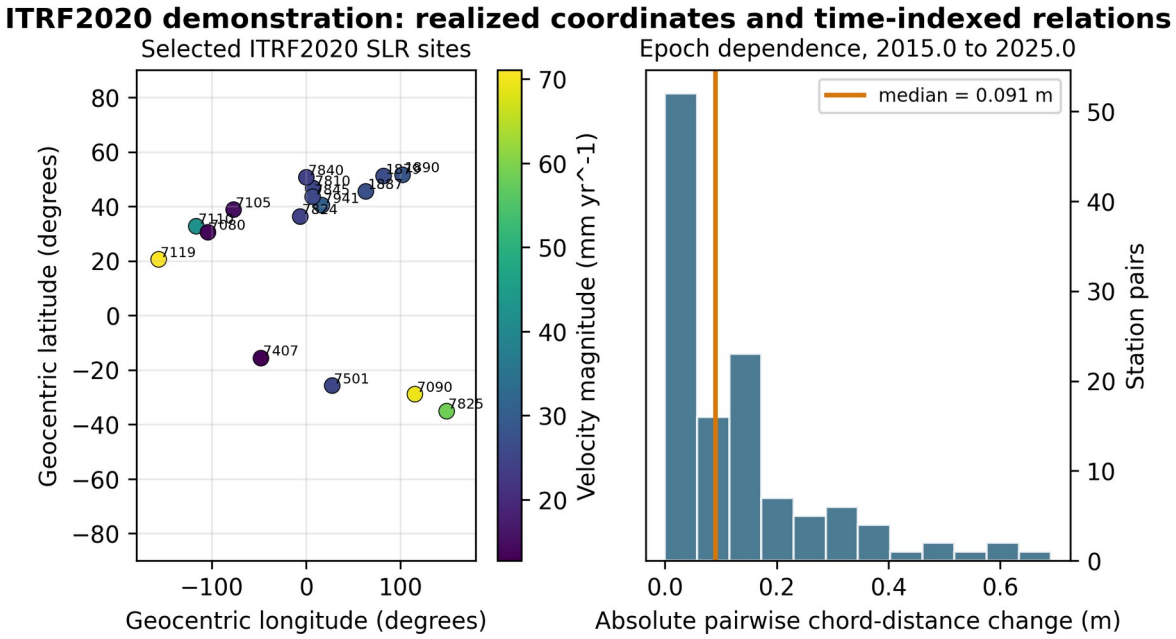


Fig. 5 ITRF2020 demonstration. The left panel shows the selected SLR sites and published velocity magnitudes. The right panel shows absolute pairwise Euclidean chord-distance changes after linear propagation from epoch 2015.0 to 2025.0; this is an epoch-sensitivity calculation, not a reflected physical realization.

6 Discussion

Why not simply store latitude, longitude, height, epoch, and the reference frame? That is the correct operational answer when the coordinate metadata, realization, transformation models, and software remain available. It is not equivalent to preserving the relation under metadata loss. Coordinates encode a chosen representation; distances encode frame-free metric information; a directed axis and physical chirality encode orientation; and redundant physical anchors provide independently testable recovery paths. Storing both coordinates and relational invariants is therefore not duplication: it permits detection of corruption, convention loss, or misapplied transformations.

The residual-invariance hierarchy complements, rather than replaces, free-network adjustment. Datum constraints choose or realize a coordinate representation within a gauge class; the present audit asks which claims are shared by every member of that class. Helmert alignment, minimum constraints, NNT/NNR conditions, and CRS metadata remain essential for operational coordinates, but they do not retroactively create a lost physical chiral witness or meridian anchor.

A plate-reconstruction model is likewise insufficient by itself. A model represents an interpretation conditioned on plate circuits, Euler rotations, paleomagnetic assumptions, hotspot or mantle anchors, and a particular absolute-longitude strategy. The model may become unavailable or superseded, and its parameters

do not replace the underlying observations, uncertainty, provenance, or entity mapping. Global plate frames seek coherence among such constraints (Torsvik et al. 2008), but geological longitude remains more model-dependent than paleolatitude (Torsvik et al. 2012; Torsvik and Cocks 2019). The appropriate archival response is to store model version and outputs together with constraints and uncertainty, not to label paleolongitude categorically impossible.

The hierarchy shows why additional anchors should be independent. Multiple coordinates derived from the same frame operation are correlated copies, not redundant physical evidence. A hybrid archive should combine labeled distances, physical chirality, directed-axis evidence, epoch/time-scale metadata, and provenance whose failure modes are not identical.

Reference-frame durability is consequently relation-specific. A terrestrial frame is precise for modern crust-fixed coordinates but depends on maintained stations and motion models. A celestial frame offers inertial orientation but requires Earth-orientation links for terrestrial directions. A spin-axis frame supports axial ordering but not a meridian. Plate-fixed, paleomagnetic, hotspot, and mantle frames extend to geological time only through additional physical assumptions. Calling any one of these frames 'permanent' without naming the preserved relation hides the actual invariance class.

A machine-readable implementation of E should preserve values, units, coordinate and time scales, identifier namespaces, covariance or samples, source checksums, model versions, and explicit statements of which symmetries remain unresolved. Reconstruction software should emit both an estimate and an ambiguity descriptor—for example, 'unique up to axial reflection'—so later users cannot mistake a chosen convention for recovered evidence.

Entity identity is a separate inverse problem. A perfect frame cannot decide whether a historical name denotes a monument, a moving plate, a city footprint, or a semantic region. Candidate mappings and their probabilities belong in I and U, and spatial reconstruction should be conditional on them.

7 Limitations

The formal results assume labeled points, a Euclidean embedding, a known metric scale, a recoverable origin for axial projection, and generic nondegenerate configurations. Real archives may contain incomplete distance graphs, deformation, topological change, scale loss, and correlated anchor errors. Surface-geodesic and non-Euclidean distance models require separate theorems. Local rigidity rank neither proves global uniqueness nor solves graph completion and identity inference.

The ITRF example demonstrates symmetry and epoch sensitivity in a modern realized network. It does not validate million-year station propagation, plate reconstructions, paleomagnetic field models, or mantle frames. Linear station velocities over ten years are used only to make the epoch dependence directly auditable.

A meridian or chiral witness can be physically defined in several ways, but recoverability, independence, uncertainty, and deformation must be assessed for each application. Paleolongitude remains model-dependent, and neither the hierarchy nor the ITRF demonstration validates a particular deep-time reconstruction mechanism.

8 Conclusions

Preserving historical direction requires more than preserving coordinates or distances. Complete distances preserve metric shape up to orthogonal gauge freedom. A directed physical axis supports axial ordering, a recoverable physical chiral witness supports signed relative longitude, and a directed meridian supports absolute angular origin. Epoch, entity identity, metric scale, uncertainty, and provenance are independent requirements.

A practical archive should therefore retain the original coordinate representation and frame metadata, relational invariants, orientation anchors, explicit epochs, uncertainty, and provenance. The correct output when information is missing is an ambiguity class or probability—not an unjustified unique reconstruction.

Declarations

Funding

[To be completed locally before submission.]

Competing interests

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Author contributions

Tatsuki Onishi: Conceptualization, methodology, validation, formal analysis, investigation, data curation, writing—original draft, writing—review and editing, visualization, supervision, and project administration.

Data availability

The persisted ITRF2020 SLR source file, its checksum ledger, the processed station selection, simulation configurations, and generated numerical results are available in the public repository at https://github.com/bougtoir/deep-time-reference-systems. The authoritative ITRF source is https://itrf.ign.fr/ftp/pub/itrf/itrf2020/ITRF2020_SLR.SSC.txt.

Code availability

The complete reproducible Python analysis and document-generation pipeline, pinned configuration, tests, and reproducibility archive are available at https://github.com/bougtoir/deep-time-reference-systems.

Ethics approval

Not applicable. The study uses synthetic geometry and public geodetic station products and involves no human participants or animals.

Generative AI disclosure

During preparation of this work, the author used the Devin web service by Cognition (no user-visible version identifier; used 24–25 September 2026). Prompts requested targeted literature-metadata verification, Python implementation and checking, deterministic simulation replication, figure/table/document generation, numerical reconciliation, and language revision. The tool assisted with those tasks but did not supply empirical source data or generate experimental images; all numerical results and display items were regenerated by the disclosed code from persisted inputs and configurations. The author reviewed and verified

the code, outputs, references, figures, and text, made the scientific and editorial decisions, and takes full responsibility for the content.

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